Normal form games

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Game Theory and Mechanism Design - Lecture 2

(This lecture heavily relies on Narahari (2014) Game Theory and Mechanism Design – chaps. 3 and 4)

Learning objectives

At the end of this lecture (and related tutorial(s)), a student should be able to:

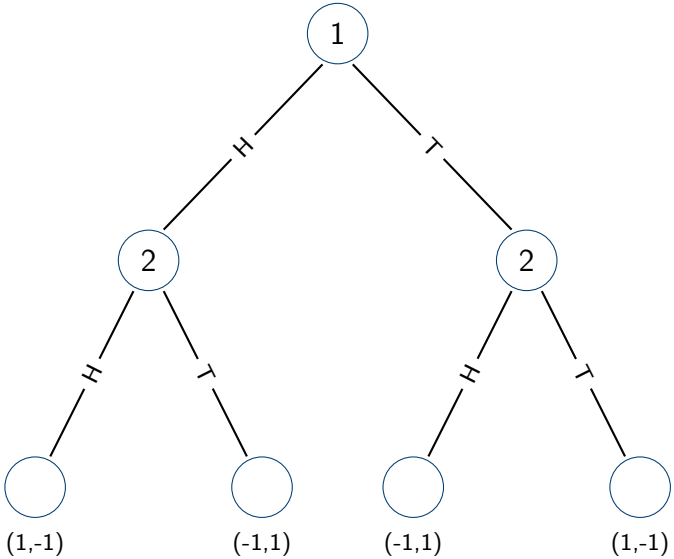
- Describe **normal form** games
- Show how to go from **extensive form** to normal form games
- Discuss the concepts defining **strategies**
- Go from description to **solution concepts**

Outline

1. Extensive form games
2. Normal form games
3. Strategies
4. Solution concepts

1. **Extensive form games**

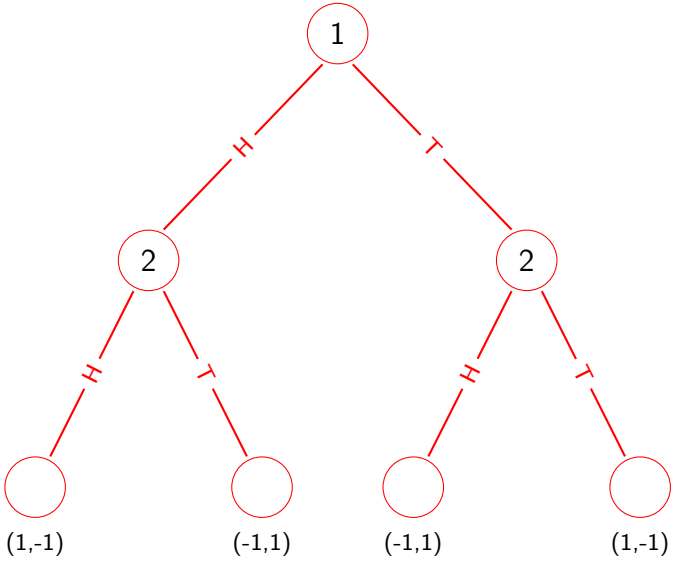
Matching pennies



Matching pennies (with observation):

- 2 players (and their coins)
- player 1 starts by choosing a side for their coin (H or T)
- player 2 observes it and chooses a side for their coin (H or T)
- If the 2 players have the same side, the player that played first gets a *pay-off* of 1, and the other one of -1
- If the 2 players have different sides, the player that played first gets a *pay-off* of -1, and the other one of 1

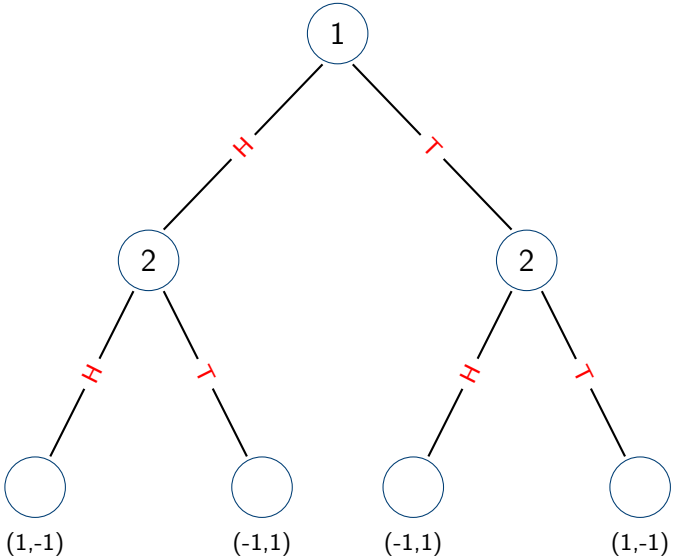
Describing an extensive form game



We have a *game tree*

It represents the overall structure of the game

Describing an extensive form game



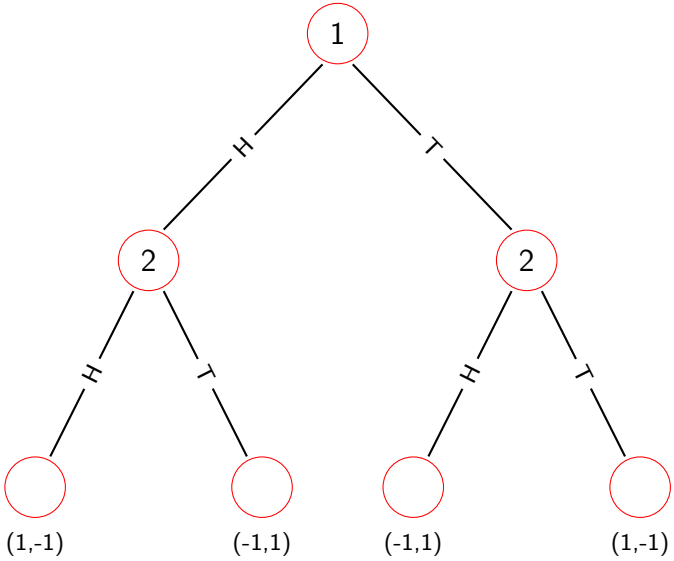
We have *actions*

They comprise the decisions available to the players.

For instance, here, the set of actions for player 1 and player 2 are:

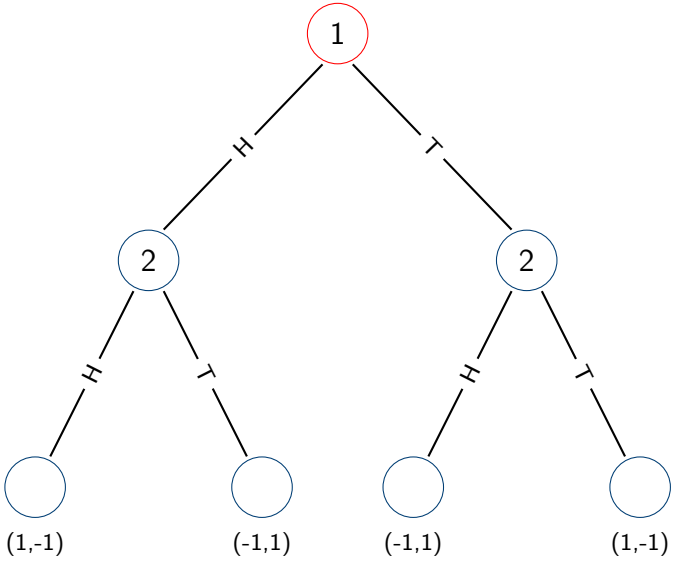
- $A_1 = \{H, T\}$
- $A_2 = \{H, T\}$

Describing an extensive form game



We have different types of *nodes*

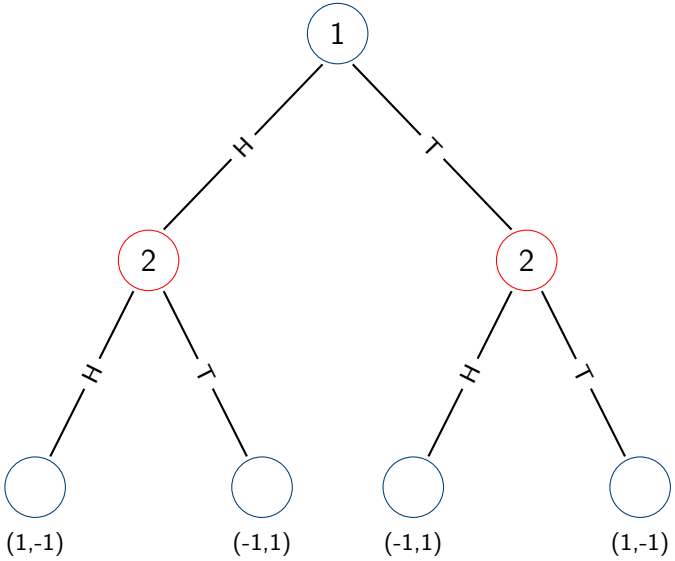
Describing an extensive form game



We have different types of nodes, i.e.,

- A *root node*

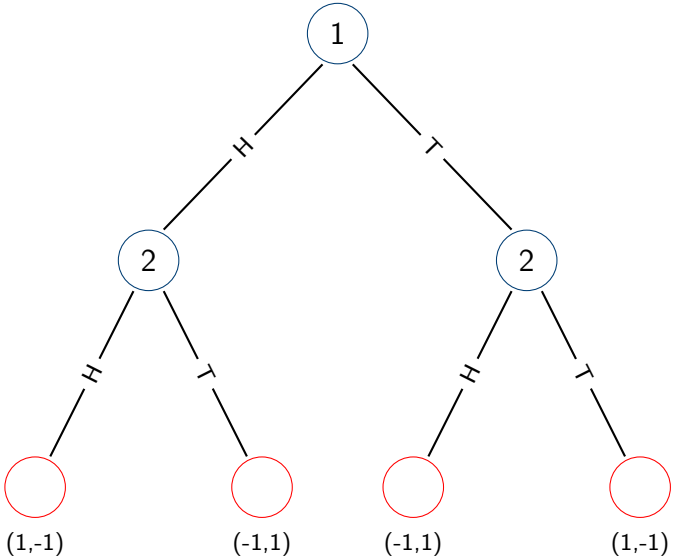
Describing an extensive form game



We have different types of nodes, i.e.,

- A root node
- *Internal (decision) nodes*

Describing an extensive form game

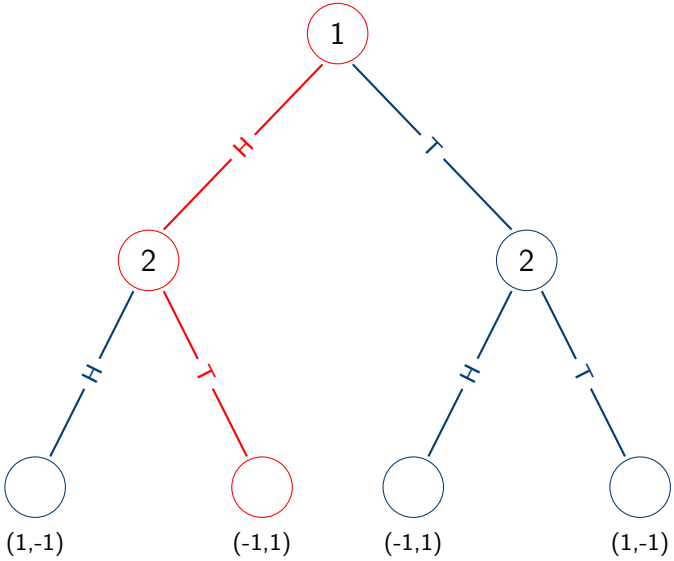


We have different types of nodes, i.e.,

- A root node
- Internal (decision) nodes
- *Terminal (outcome) nodes*
  (also referred to as *leaf nodes*)

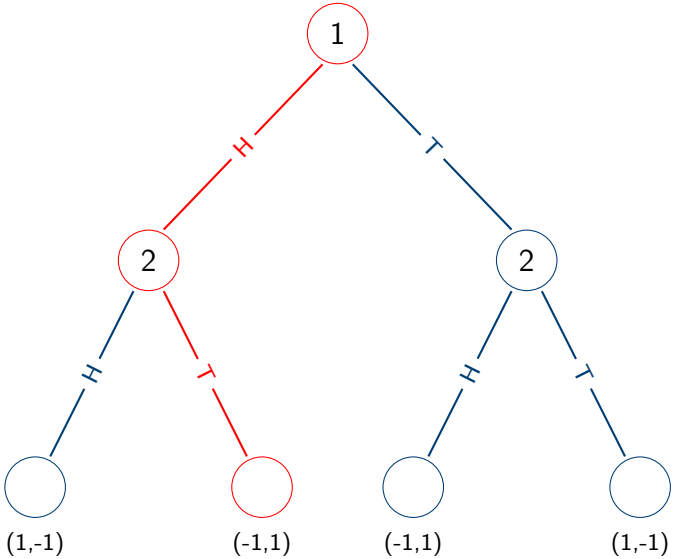
Terminal nodes also show the *pay-offs* (or, utility) for all players, if reaching those nodes

Describing an extensive form game



Any sequence between the root node and a terminal node is referred to as a *terminal history* (or *path of play*)

Describing an extensive form game

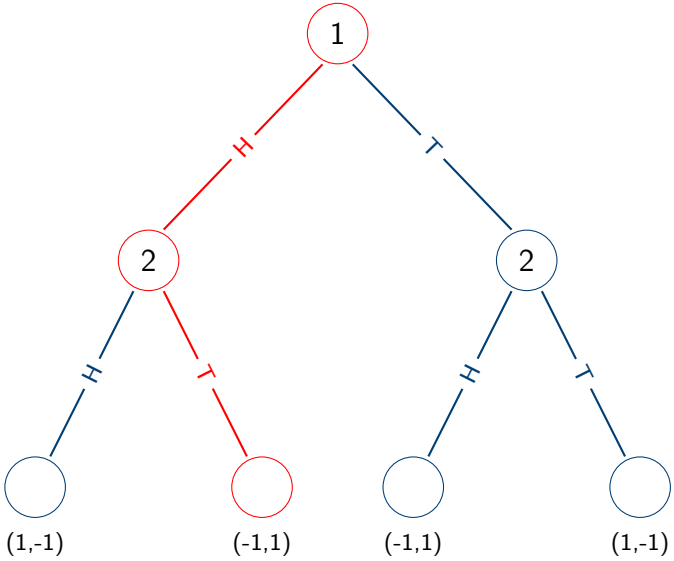


Any sequence between the root node and a terminal node is referred to as a *terminal history* (or *path of play*)

All terminal histories are gathered within the set $\mathbb{H}$, e.g.

$$\mathbb{H} = \{(H, H), (H, T), (T, H), (T, T)\}$$

Describing an extensive form game



Any sequence between the root node and a terminal node is referred to as a *terminal history* (or *path of play*)

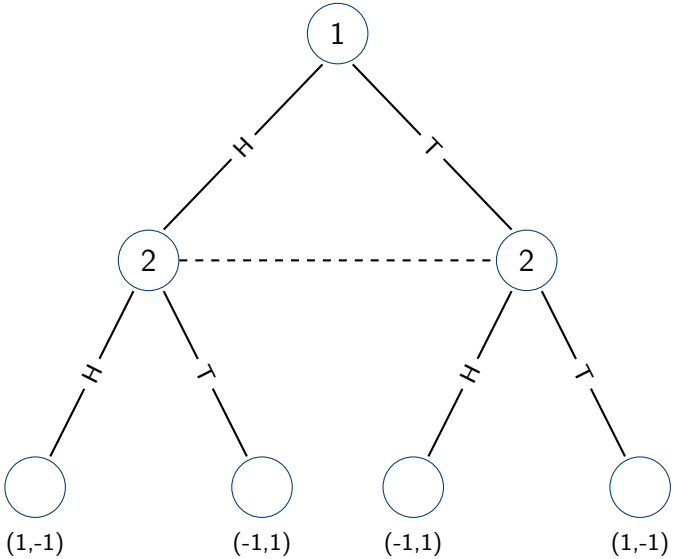
All terminal histories are gathered within the set $\mathbb{H}$, e.g.

$$\mathbb{H} = \{(H, H), (H, T), (T, H), (T, T)\}$$

For each terminal history, all players perceive a given pay-off (or utility), i.e., here,

- $u_1(H, T) = -1$
- $u_2(H, T) = 1$

Matching pennies

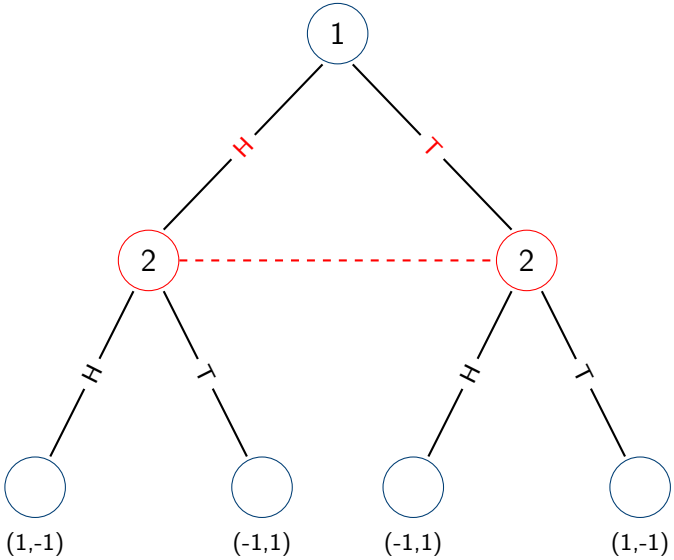


Matching pennies (without observation):

- Same setup as before, except that...
- player 2 does not see the coin of player 1 before to make their decision

(It gets a bit more interesting!)

Describing an extensive form game



The decision nodes for player 2 comprises an **information set**

Definition (Information set)

An *information set* of a player is a set of their own decision nodes (and of their history) that are indistinguishable to them.

Here,

- player 1 has a single information set $\{\emptyset\}$
- player 2 has a single information set $\{H, T\}$

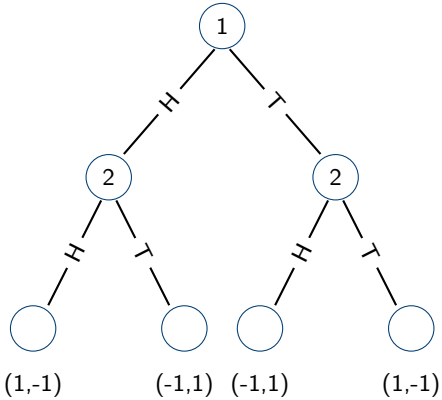
Write $\mathbb{I}_1 = \{\{\emptyset\}\}$ and $\mathbb{I}_2 = \{\{H, T\}\}$ their sets of informations sets.

(for *matching pennies with observation*, player 2 has 2 distinct information sets, i.e., $\mathbb{I}_2 = \{\{H\}, \{T\}\}$)

Definition (Perfect and imperfect information games)

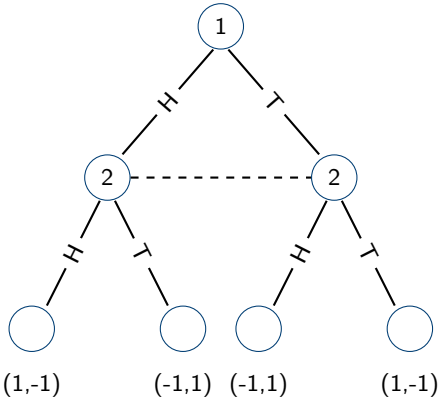
An extensive form game is said to be with *perfect information* if the information sets of all players are singletons only. It is with *imperfect information* if that is not the case.

Perfect information



$\mathbb{I}_1 = \{\{\emptyset\}\}, \ \mathbb{I}_2 = \{\{H\}, \{T\}\}$

Imperfect information



$\mathbb{I}_1 = \{\{\emptyset\}\}, \ \mathbb{I}_2 = \{\{H, T\}\}$

Defining strategies

Definition (Strategy)

A strategy s_i of player i is a mapping $s_i : \mathbb{I}_1 \to A_i$, such that $s_i(J) \in C(J)$, $\forall J \in \mathbb{I}_i$.
Given an information set $J \in \mathbb{I}_i$, $C(J) \subseteq A_i$ is the set of all actions possible to player i based on the information set J.

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In this case,

- $A_1 = A_2 = \{H, T\}$
- $\mathbb{I}_1 = \{\{\emptyset\}\}$, $\mathbb{I}_2 = \{\{H, T\}\}$

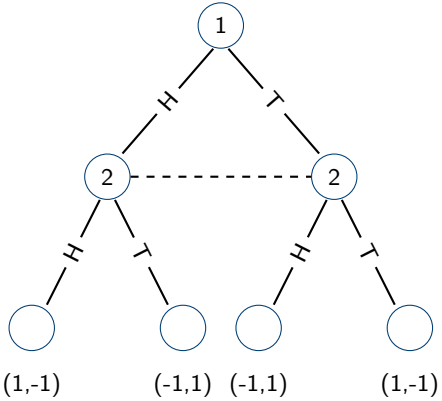
player 1 has 2 strategies:

- $s_{11} : \{\emptyset\} \to H$
- $s_{12} : \{\emptyset\} \to T$

player 2 has also 2 strategies:

- $s_{21} : \{\{H, T\}\} \to H$
- $s_{22} : \{\{H, T\}\} \to T$

Imperfect information



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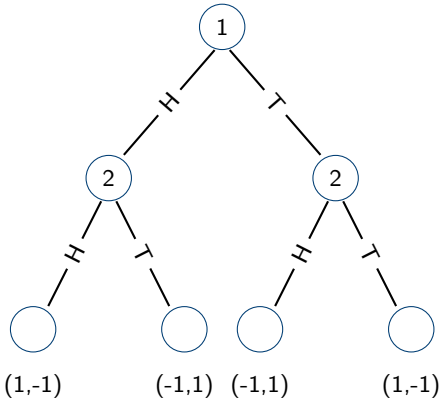
Perfect information



In this case,

- $A_1 = A_2 = \{H, T\}$
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player 1 has 2 strategies:

- $s_{11} : \{\emptyset\} \to H$
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player 2 now has 4 strategies:

- $s_{21} : \{H\} \to H,\ \{T\} \to H$
- $s_{22} : \{H\} \to H,\ \{T\} \to T$
- $s_{23} : \{H\} \to T,\ \{T\} \to H$
- $s_{24} : \{H\} \to T,\ \{T\} \to T$

2. **Normal form games**

Normal form games are also strategic form games

Theorem

Any extensive form game can be uniquely defined based on strategies and related pay-offs. Hence, any extensive form game can be transformed into an equivalent strategic form game (also referred to as normal form game).

- Each player i has their strategy set S_i
- Their combination $S = \prod_i S_i$ define the set of *strategy profiles* for the game

If looking again at the games of matching pennies, we obtain a *matrix representation* (or *matrix form*):

Perfect information

- $S_1 = \{s_{11}, s_{12}\}$, $S_2 = \{s_{21}, s_{22}, s_{23}, s_{24}\}$
- $S_1 \times S_2 = \{(s_{11}, s_{21}), (s_{11}, s_{22}), (s_{11}, s_{23}), (s_{11}, s_{24}), (s_{12}, s_{21}), (s_{12}, s_{22}), (s_{12}, s_{23}), (s_{12}, s_{24})\}$

player 2

player 1		s_{21}	s_{22}	s_{23}	s_{24}
	s_{11}	$1, -1$	$1, -1$	$-1, 1$	$-1, 1$
	s_{12}	$-1, 1$	$1, -1$	$-1, 1$	$1, -1$

Imperfect information

- $S_1 = \{s_{11}, s_{12}\}$, $S_2 = \{s_{21}, s_{22}\}$
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player 2

player 1		s_{21}	s_{22}
	s_{11}	$1, -1$	$-1, 1$
	s_{12}	$-1, 1$	$1, -1$

Definition (Normal form game)

A *normal form game* Γ is a tuple $\langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$ where

- $N = \{1, \ldots, n\}$ is a set of players
- S_i is the strategy set of player i
- u_i are the utility (or pay-off) function, for player i, such that $u_i : S_1 \times \ldots \times S_n \to \mathbb{R}$

The way the game is played is then:

- Players **simultaneously** choose a given strategy from their strategy set, which they apply
- The combination of these strategies define a **strategy profile**
- The strategy profile yields a specific **outcome**
- Each player gets a **pay-off** from this outcome

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Important notations - as it is convenient to take the point of view of a single player (say, player i):

- write s_i the strategy picked by player i, and s_{-i} the strategies picked by all other players, i.e. $s_{-i} = (s_1, \ldots, s_{i-1}, s_{i+1}, \ldots, s_n)$
- similarly, we can oppose the strategy set S_i of player i to the strategy set S_{-i} of all other players, i.e. $S_{-i} = S_1 \times \ldots \times S_{i-1} \times S_{i+1} \times S_n$
- a strategy profile is then simply (s_i, s_{-i})

Example: Prisoner dilemma

It might be the most famous application example from game theory!

- 2 individuals are arrested for a crime (player 1 / player 2)
- They are both interrogated individually and without possibility of communication between them
- The interrogator tells each of them privately that
  - if none of them confess, they will both get 2 years in jail
  - if confessing for both of them, while the other one does not do so, the one confessing will get a single year in jail, and the other one 10 years
  - if confessing for both of them, and they both do so, they will both only get 4 years in jail
- Write the 2 possible actions C (**C**onfess) and NC (**N**ot **C**onfess)

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In **matrix form**:

		player 2	
		C	NC
player 1	C	$-4, -4$	$-1, -10$
	NC	$-10, -1$	$-2, -2$

3. **Strategies**

Pure strategy

Definition (Pure strategy)

A pure strategy for player i is a deterministic plan of action (i.e., for a given information set as input, player i will always use the same action).
The set of all pure strategies for player i is denoted by S_i.
A profile of pure strategies $s = (s_1, \ldots, s_n)$ describes a combination of individual pure strategies for the n players in the games.

For the **prisoner dilemma** example:

- the set of pure strategies, for both players, is $S_1 = S_2 = \{C, NC\}$
- a *pure strategy* for player 1 can be to confess (C)
- a *pure strategy* for player 2 can be not to confess (NC)
- the resulting strategy profile is $(s_1, s_2) = (C, NC)$
- the pay-offs for players 1 and 2 are $u_1 = -1$ and $u_2 = -10$, respectively.

Best response strategy

Eventually, we want to analyse these games... so a fair question is **"what will a given player do?"**

Definition (Best response strategy)

Given a strategic form game $\Gamma = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$ and a strategy profile $s_{-i} \in S_{-i}$ (for the other players), the best response strategy is player i is given by

$$s_i = \underset{s}{\operatorname{argmax}}\ u(s, s_{-i})$$

i.e., necessarily leading to $u(s_i, s_{-i}) \geq u(s_i', s_{-i})$, $\forall s_i' \neq s_i$.

For the **prisoner dilemma** example, assuming you are **player 1**:

- *if you knew* that player 2 was not going to confess, your best response strategy is to **confess** (C)
- *if you knew* that player 2 was going to confess, your best response strategy is to **confess** (C)

4. **Solution concepts**

What solution concept?

And, more generally, we want to be able to **"predict"** what the overall outcome may be!

Definition (Equilibrium)

For a given finite normal game (i.e., with a finite number of players and a finite number of strategies), an *equilibrium* is to be seen as a prediction of a likely outcome, in terms of the strategy profile that will prevail, and pay-offs for the players involved.

What would you see as an equilibrium for the **prisoner dilemma**? and why?

What solution concept?

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What would you see as an equilibrium for the **prisoner dilemma**? and why?

- for player 1, the best response strategy is to **confess**, whatever happens
- for player 2, the best response strategy is to **confess**, whatever happens

So... it looks like the equilibrium will be the strategy profile (C,C), and with pay-offs (-4,-4).

What solution concept?

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What would you see as an equilibrium for the **prisoner dilemma**? and why?

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- for player 2, the best response strategy is to **confess**, whatever happens

So... it looks like the equilibrium will be the strategy profile (C,C), and with pay-offs (-4,-4).

Does that correspond to the **best overall outcome**, in your opinion?

Do we always have an equilibrium?

The **"battle of the sexes"** is another classic game (originally introduced by Luce and Raiffa in 1957)

- A couple (2 players, Alex and Parker) wants to go out in the evening
- They do not communicate and will directly meet up at the place
- They obviously prefer ending up together
- The 2 options are
  - Opera, preferred by Alex, but not by Parker
  - Football game, preferred by Parker, but not by Alex
- Write the 2 possible actions O (**O**pera) and F (**F**ootball)

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In **matrix form**:

		Parker	
		O	F
Alex	O	$2, 1$	$0, 0$
	F	$0, 0$	$1, 2$

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In **matrix form**:

		Parker	
		O	F
Alex	O	$2, 1$	$0, 0$
	F	$0, 0$	$1, 2$

What are their best response strategies, and the potential equilibrium?

Wrapping up

- **Normal form games** can represent many practical problems in coordinating multi-agent systems
- The concepts of **strategies** and **equilibria** are key to game theory
- **Not all games** have an equilibrium (**existence** property)
- Some games have **multiple equilibria** (**uniqueness** property)

Further readings (/lecture notes for this topic):

Narahari Y (2014) *Game Theory and Mechanism Design* - Chaps 3 and 4 (available with this lecture on Blackboard Learn)

Thanks for your attention!